Caitlyn McLuskie, Josias Cruz

ENGL-2116

3/30/2020

Use of Virtual Personal Assistants to Combat Mental Illness in Social Isolation

Introduction

Jane Doe, a friend of mine who asked to remain anonymous as this example, suffers from both severe anxiety and moderate depression. She is terrified of getting sick at the best of times, so the current spread of coronavirus has her nearing a panic attack multiple times a day. Jane's increased hypochondria has made her next to incapable of leaving her home. This has left Jane feeling more isolated than usual. Like many mentally ill individuals now, Jane is in a precarious position: miss out on the help she desperately needs, or face a world that will send her spiraling before she even makes it to her therapist's office. Artificial Intelligence provides a third option. Jane can open up an app on her phone, safe in her heavily sanitized home, and get access to breathing exercises, coping strategies, and a hotline for emergencies.

In our current climate of social distancing, many face the risk of deteriorating mental health. Even for those less affected than Jane, loneliness and isolation are real threats to one's health. Artificial Intelligence can provide a much-needed solution for those who are struggling. When utilized efficiently, automated communication allows people to seek help without fear of judgment and receive reliable advice. Traditional therapy is often intimidating and unreachable to those who need it most. With artificial intelligence, there are no office hours, scheduling conflicts, or transportation costs.

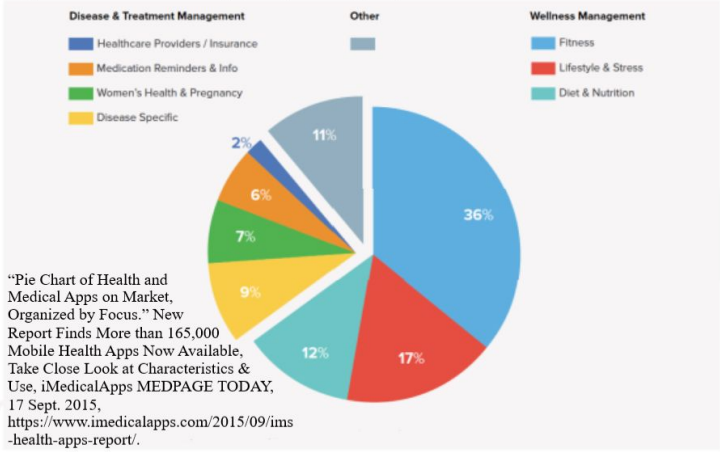
Though a mental health application is not comparable to the treatment one gets from a licensed psychotherapist, it can be used in a multitude of ways. In-person cognitive behavioral therapy, the type globally recommended as the primary course of action in treating mental illness, is often intimidating. Patients with anxiety, for example, are often not able to call and schedule appointments. Depressed patients may not be able to pull themselves out of an episode to talk to someone about treatment. These barriers to receiving help can make getting better seem like an impossible task. Nowadays, most people in Western Society are "plugged in". We are on some form of technology nearly every waking hour. Regardless of whether you believe that is a detriment or benefit to society as a whole, it can certainly be a benefit for individuals who are struggling.

People in the chronic illness community use a term: "spoons" to communicate the tasks they are physically able to perform on a given day. The metaphor is as follows: each person gets a certain number of "spoons" per day and when they run out, they cannot do anything else. This way of explaining fatigue has been loosely adopted by some mentally ill. For someone who is depressed, they may only get ten "spoons" a day. Eating breakfast takes one "spoon" and taking a shower takes three "spoons", but therapy may take seven "spoons". By the time they get to the office and back home, that person is exhausted and they cannot do anything else that day. This could discourage them from going to therapy as often as they may need to. Checking in with a personal assistant who cares about your wellbeing, on the other hand, may not even take a "spoon". For many individuals, it is an easier way to keep up with your wellbeing between therapeutic sessions. Therapists have also begun to ask patients to use a mental health

management application so they can see trends and pick up on things their patients may not have explicitly told them.

Technology-Based Health Management More individuals are now taking advantage of technology to manage health concerns. As seen in Figure A (below), there are many uses for a health management application. The majority of users as of 2015 were using these types of applications to keep track of physical and mental wellness. With the integration of artificially intelligent software, these apps will be more effective. They will be able to encourage the user to meet fitness goals, take medicines, and keep track of disease-specific needs. In addition, artificial intelligence capabilities provide a personalized approach towards healthcare without the cost that comes with daily one-on-one care.

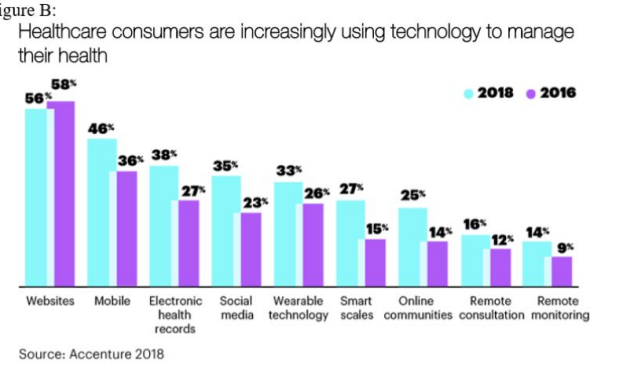
Figure A: mHealth Apps by Category 2015



"Pie Chart of Health and Medical Apps on Market, Organized by Focus." New Report Finds More than 165,000 Mobile Health Apps Now Available, Take Close Look at Characteristics & Use, iMedicalApps MEDPAGE TODAY, 17 Sept. 2015, https://www.imedicalapps.com/2015/09/ims-health-apps-report/.

In addition to application-based health management, there is a growing number of users who are taking advantage of alternate mediums to do the same. According to Figure B (right), while applications and other mobile technologies have grown in consumer use, so have many other ways for users to manage their wellbeing.

Figure B:



"Graph of Various Technological Mediums Used to Manage Health." The Rise of MHealth Apps: A Market Snapshot, Liquid State, 26 Mar. 2018, https://liquid-state.com/mhealth-apps-market-snapshot/.

Scope

For the purposes of this project, the scope will be to create an emotionally intelligent virtual personal assistant (VPA) that is capable of handling the emotions of a human being under social distancing protocols. The emotionally intelligent VPA is intended to be developed with both voice capability and text capability communication channels. This is done with the purpose of boosting the ability of a person to create an emotional bond with the personal assistant. (Looije). The user can then develop a trust with it so that they can easily seek help from it without hesitation. Both options offer the widest usability throughout various types of abled populations.

The emotionally intelligent VPA will be based on Western American social customs. The language set will be English as well as the regional preferences of the social norms of the United States. The emotionally intelligent VPA will not take into account some forms of banter. For example, the cases where a person might say something really mean in a positive voice, as that would throw the user off guard and it adds a layer of complexity.

The main problems this emotionally intelligent VPA will seek to resolve are problems with depression and anxiety. Both of these are worsened when social distancing and isolation increase. The emotionally intelligent VPA will be able to handle most minor cases of depression and anxiety in this case, similar to the cases being able to be solved by friends in the initial stages. The emotionally intelligent VPA will have to determine whether a person or user is suicidal.

Due to the complexity of suicide, the emotionally intelligent VPA will not be able to handle cases of suicidal depression or anxiety. Both of which could happen at any time during depressive and anxious cycles. When such cases are determined to be present the emotionally intelligent VPA will connect in a suicide hotline and will enter a mediation process. The mediation process will seek to acclimate the user for a transfer between the emotionally intelligent VPA to a certified trained crisis/suicide management human agent.

Methodology

Developing Emotional Intelligence In order to teach emotional intelligence to the VPA, there has to be a baseline. As such, a massive dataset would have to be created that details a multitude of basic social scenarios and expected reactions. The data set will take into account basic greetings and expected responses and rate them based on desirability. There will have to be data sets based on regional differences as well. Surveys of each region will have to be accumulated. Once the data is gathered, the emotionally intelligent VPA will have to begin to learn the expected behaviors and various variations.

Personality profiles for the VPA can be built based on the desirability ratings and the VPA will be taught how to switch between personalities again via another set of ratings for personality switches. The key part of this is the analysis that will have to be made in part by the VPA to interpret the user feedback to determine the best next course of action. Because the VPA will have ratings based on response desirability, the logic can be reversed to be considered in response to feedback. For example, if I know that when I'm serious the expected behavior is I

don't make jokes, then if someone asks me how to be serious I would suggest not making jokes. That suggestion is born out of my notion that making jokes is not desired when serious.

Voice-based System The voice-based system will be based entirely on voice recognition algorithms. This system will take in voice input to determine frequencies and speech pattern recognition. With these patterns in place, we can begin to develop a baseline for the user. And making a correlational model between voice inputs and the personality inputs we can create moods that the emotionally intelligent model can use to determine the mood of the user.

With a system in place that can determine mood based on voice the emotionally intelligent VPA will also be able to match a voice pattern that reflects the mood that it's trying to portray. This is the same thing as the personality matching based on desirability but now adding a voice layer to the complexity. This, in turn, will allow the interaction to feel more natural and will entice the user to think of the emotionally intelligent VPA as less of a program and more of an assistant. Regional variations for pronunciation will have to be taken into account through additional data sets of pronunciations each given correlational ratings based on the desirability of the regional understanding. Should a user expect the program to speak in a southern accent the program will look at the datasets and pull from the expected responses.

Text-Based System The text-based system will simply be an application that will analyze the words being inputted and return the desired response. By this point, the datasets that had been used to train are the ones that are used to develop these responses too. The user design in this system will carry the burden of creating an environment that the user will want to engage in. Color scheme choices should be neutral. To make this system more dynamic there could be a

development of different color schemes, such as warm and cool that transition based on the user feedback.

For that functionality to be implemented there would be a correlational relationship between moods and colors and then it could be launched since the text will be used as the basis for the emotionally intelligent VPA to determine the mood and thus be able to transition to colors that aid the user in their crisis. With the color transition in place, and the ability to be able to appropriately respond to the user's needs the text-based system would be complete.

Hybrid System Up until now the Voice and Text systems have worked independently of each other. Should resources be available the best approach is to keep them both separate as that will produce a higher quality application? But there is a possibility of merging the two systems together. Everything could be built based on the Text-Based System and then add a voice layer over. The reason why this is not recommended is the voice layer would not provide the dynamic mood changing and capacity to portray emotions that the separate voice system could thus reduce the likeliness that the user could develop a bond.

Suicide Model The Suicide Model is built in order for the emotionally intelligent VPA to determine when it's no longer able to help a person in social distancing or isolation with their depressive or anxious feelings. For this function to execute properly there are a couple of data models that need to be in place. Once a user begins to use excessive negative language towards itself the app will track the duration. There will need to be a data set build on escalations of self-threats and also a data set on definitions of existentialism. Should both be present that will fall outside of the scope of this emotionally intelligent VPA and will proceed to the transition mode.

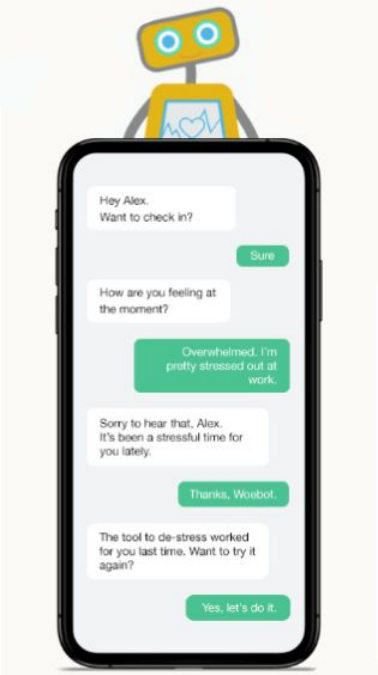
Transition Model The transitional model will work to aid the escaping of scope and the transition of a user who is in crisis to the appropriate human resources. The emotionally intelligent VPA will work on maintaining the mental state of the user and slowly work on reserving the situation, meanwhile contacting a suicide hotline. While the emotionally intelligent VPA is working on stabilizing the condition the hotline assistant will be listening in for a short period of time and then the transition will end with a handoff of the user between the emotionally intelligent VPA and the suicide hotline assistant.

Expected Behavior At the end result, the flow of the application will follow this case. A user is socially distancing or isolated due to life circumstances. The user initiates the program using a voice command. The user then begins interacting with the application, usually starting with simple tasks at first and gradually increasing the complexity as dependent on the level of trust between the user and the emotionally intelligent VPA.

Once the user feels their problem is solved, then the interaction would end. In the case that the user begins to question their existentialism, or begins to have negative self-language to the point where the emotionally intelligent VPA determines that it's excessive then the transition model is triggered and the user is passed off to a better-qualified environment. The text-based system will function in a similar fashion but will be based on an application platform.

Figure C (top left of the next page) demonstrates how a text-based aid system could function. This is a similar system developed by Woebot but this would be implemented as per the text-based system specifications outlined in the section before. The user would begin chatting with the emotionally intelligent VPA and express their emotions and the interaction would take place. The emotionally intelligent VPA in the background would analyze the mood to determine

Figure C:



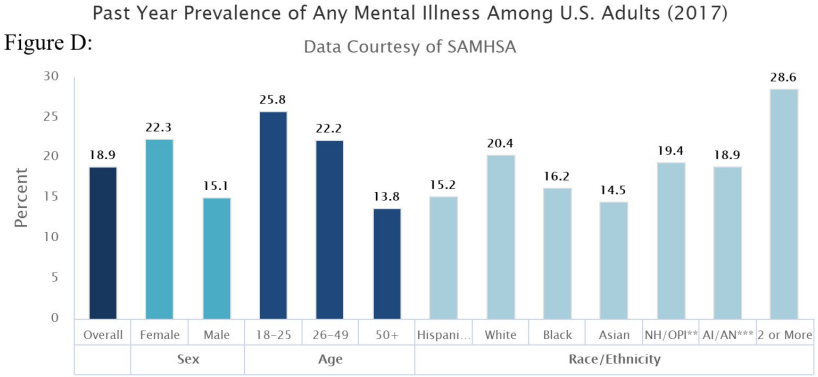
"Animated Image Made Static of an Example Conversation between Woebot and User." Woebot Homepage , Woebot, https://woebot.io/.

the color scheme of the display and would also trigger the transition model should it deem necessary under the same conditions that the voice-activated emotionally intelligent VPA would.

Impact

In 2017 alone, eighteen percent of all adults in the United States had one or more mental illnesses (see Figure D, right). Of that, nearly twenty-six percent were eighteen to twenty-five years old. That is

Figure D:

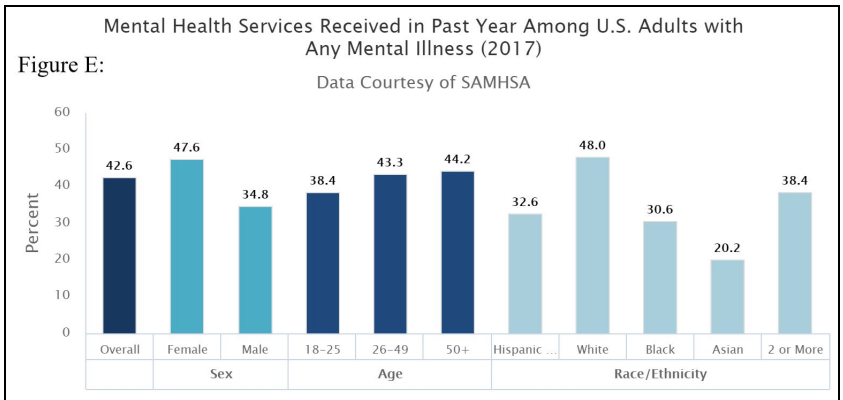


"Bar Graph of Percentages of U.S. Adults Categorized by Sex, Age, and Race/Ethnicity, Who Have Any Mental Illness. National Institute of Mental Health, https://www.nimh.nih.gov/health/statistics/mental-illness.shtml.

the age group most likely to need emotionally intelligent virtual personal assistants, as well as the most likely to benefit strongly from them. The high prevalence of younger adults diagnosed with mental illness makes sense due to the millennial generation's increasing positive perception of the effectiveness of therapy.

In addition to the increasing prevalence of diagnosed mental illness in American adults, there is also the issue of how few of those individuals are receiving mental health treatment and services. According to Figure E (right), less than half of all U.S. adults with any mental illness are actually receiving services to better their mental wellbeing. For males, that percentage is even lower. Access to an affordable, benign treatment option for minor mental illness is revolutionary in the lives of its users.

Figure E:



"Bar Graph of Percentages of U.S. Adults Who Received Mental Health Care for Any Mental Illness, as Categorized by Race/Ethnicity." Mental Illness, National Institute of Mental Health, https://www.nimh.nih.gov/health/statistics/mental-

Reader, imagine for a moment a world where no one feels alone. Everyone has a miniature therapist in their pocket who can help them through the bad times. This emotionally intelligent virtual personal assistant has the potential to vastly improve the quality of life for individuals suffering from minor mental illnesses. It can provide a helping hand to therapists when treating their patients. It can even save a life by referring an at-risk user to trained crisis management professionals. Now, Reader, doesn't that sound like a world you'd like to live in?

Visuals Cited List

Figure A:
"Pie Chart of Health and Medical Apps on Market, Organized by Focus." *New Report Finds More than 165,000 Mobile Health Apps Now Available, Take Close Look at Characteristics & Use*, iMedicalApps MEDPAGE TODAY, 17 Sept. 2015, https://www.imedicalapps.com/2015/09/ims-health-apps-report/.

Figure B:
"Graph of Various Technological Mediums Used to Manage Health." *The Rise of MHealth Apps: A Market Snapshot*, Liquid State, 26 Mar. 2018, https://liquid-state.com/mhealth-apps-market-snapshot/.

Figure C:
"Animated Image Made Static of an Example Conversation between Woebot and User." *Woebot Homepage* , Woebot, https://woebot.io/.

Figure D:
"Bar Graph of Percentages of U.S. Adults Categorized by Sex, Age, and Race/Ethnicity, Who Have Any Mental Illness." *Mental Illness*, National Institute of Mental Health, https://www.nimh.nih.gov/health/statistics/mental-illness.shtml.

Figure E:
"Bar Graph of Percentages of U.S. Adults Who Received Mental Health Care for Any Mental Illness, as Categorized by Sex, Age, and Race/Ethnicity." *Mental Illness*, National Institute of ental Health, https://www.nimh.nih.gov/health/statistics/mental-illness.shtml.

Annotated Bibliography

Coplan, Robert J. *The Handbook of Solitude Psychological Perspectives on Social Isolation, Social Withdrawal, and Being Alone*. 1st ed., Blackwell Publ., 2014.

This book is a collection of views on isolation in its many forms. It explores all aspects of isolation through a human holistic perspective. The source is a collection of various articles written by other authors regarding isolation, in itself, it appears less worthy but its diversity is the strength of the source. This source is worth using as a standard and will plan to be used as a base level analysis of isolation and its nature.

Dahlke, Deborah V, and Marcia G Ory. "Emerging Opportunities and Challenges in Optimal Aging with Virtual Personal Assistants." *Public Policy & Aging Report*, vol. 27, no. 2, 29 July 2017, pp. 68–73., doi:https://doi.org/10.1093/ppar/prx004.

This article deals with the nuances of voice-activated personal assistants (VAPAs) and their interactions with elderly people. It scopes from accessibility to the possibility of its effectiveness in the optimal progression of humans. Dahlke and Ory have an extensive grasp over the utilization of VAPAs throughout the aging lifecycle. This source will be used as a basis on what the technology methodology will be concerned with and will bring issues that need to be mitigated.

Dautenhahn, Kerstin. "Getting to Know Each Other—Artificial Social Intelligence for Autonomous Robots." *Robotics and Autonomous Systems*, vol. 16, no. 2-4, Dec. 1995, pp. 333–356., DOI:10.1016/0921-8890(95)00054-2.

This experiment explores a way of creating social intelligence between robots. This experiment explores the breakdown of social intelligence into components that can be transferred to computer logic. The experiment goes thoroughly for robot-to-robot interactions and social intelligence among them and the building of such, and even expands it to it's usability to humans as well. Overall this is a great source for our proposal due to the breakdown of the learning of emotional intelligence.

Looije, Rosemarijn, et al. "Persuasive Robotic Assistant for Health Self-Management of Older Adults: Design and Evaluation of Social Behaviors." *International Journal of Human-Computer Studies*, vol. 68, no. 6, June 2010, pp. 386–397., DOI:10.1016/j.ijhcs.2009.08.007.

This study discusses the use of computer-based personal assistance to improve daily health self-management in older adults with obesity or diabetes. It was found that older adults respond

more positively to effective persuasion characterized by the assistant's expression of social and empathetic behaviors. Participants in this study preferred a virtual or physical character as opposed to a text-based interface, as they trusted the advice of the computer-based assistants when they were more human-like.

Yang, Heetae, and Hwansoo Lee. "Understanding User Behavior of Virtual Personal Assistant Devices." *Information Systems and e-Business Management*, vol. 17, no. 1, 2018, pp. 65–87., DOI:10.1007/s10257-018-0375-1.

This study investigated the factors that lead to strong usage intention with virtual personal assistant devices. The facts found to be the most impactful were perceived usefulness and enjoyment. In layman's terms, a virtual personal assistant was found to be used most like the creators intended when the user-perceived it to add usefulness and enjoyment to their lives.

Fitzpatrick, Kathleen K, et al. "Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial." *JMIR Mental Health*, JMIR Publications Inc., Toronto, Canada, 6 June 2017, mental.jmir.org/2017/2/e19/.

This study set out to determine the feasibility, acceptability, and efficiency of an app that is fully automated to converse with users and provide self-help. It was tested on college students self-identified as symptomatic of anxiety and depression. The app, called Woebot, provides self-help tools and advice to users that follows traditional cognitive-behavioral therapeutic practices. It was found to significantly reduce symptoms of anxiety and depression during the study period based on participant comments. Additionally, participants were found to be more accepting of a conversational agent as opposed to traditional therapy.